

MODEL RESEARCH DOCUMENT

Project Title

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Introduction

This document evaluates machine learning, deep learning, natural language processing, retrieval, and generative AI approaches suitable for an AI-Powered Workforce Analytics & Talent Intelligence Dashboard. The selected models support employee attrition prediction, workforce planning, skill-gap identification, talent segmentation, anomaly detection, resume and feedback analysis, semantic search, conversational analytics, and AI-assisted recommendations. The comparison highlights their primary use cases, advantages, and indicative expected performance for workforce analytics applications.

Additional Models and Technologies

Model / Technology	Use Case	Advantages	Expected Performance
Logistic Regression	Attrition baseline	Simple and highly interpretable	80–88%
Random Forest	Attrition and retention risk	Robust nonlinear modeling	88–94%
XGBoost	High-risk employee prediction	Strong tabular performance	90–96%
LightGBM	Large workforce datasets	Fast and memory efficient	90–96%
CatBoost	Categorical HR data	Native categorical handling	91–97%
Explainable Boosting Machine	Transparent risk scoring	Interpretable nonlinear effects	88–94%
Cox Proportional Hazards	Time-to-attrition analysis	Estimates event risk over time	75–88%
Random Survival Forest	Nonlinear time-to-exit risk	Flexible survival modeling	78–90%
K-Means	Talent segmentation	Simple employee grouping	High Accuracy
HDBSCAN	Irregular workforce segments	Finds clusters and noise	High Precision
Isolation Forest	Workforce anomaly detection	Scalable outlier detection	High Precision
Autoencoder	Complex anomaly patterns	Learns hidden representations	High Accuracy
Sentence-BERT	Skill and profile similarity	Semantic text embeddings	88–95%
E5 / BGE Embeddings	RAG workforce retrieval	Strong semantic retrieval	85–95%
DeBERTa	Skill and entity extraction	Strong contextual understanding	90–96%
Longformer	Long HR document analysis	Handles long-context text	88–95%
FLAN-T5	HR Q&A; and summaries	Instruction-tuned generation	High Accuracy
Llama-family Instruct	Conversational workforce assistant	Flexible enterprise assistant	High Performance
RAG Pipeline	Evidence-grounded HR Q&A;	Improves grounded responses	High Precision
SHAP	Prediction explanation	Local and global interpretability	High Explainability

Recommended Model Architecture

A practical implementation should use a layered architecture rather than a single model. For attrition prediction, Logistic Regression can serve as an interpretable baseline and can be compared with Random Forest, XGBoost, LightGBM, and CatBoost. For time-aware retention analysis, Cox Proportional Hazards or Random Survival Forest can be applied. For workforce segmentation, K-Means or HDBSCAN can identify employee groups, while Isolation Forest can detect unusual workforce patterns. For skill intelligence and semantic matching, Sentence-BERT or E5/BGE embeddings can be stored in a vector index. A Retrieval-Augmented Generation pipeline can retrieve relevant workforce evidence before an instruction-tuned language model generates answers, summaries, and recommendations. SHAP can support explanation of high-impact risk predictions.

Final Recommendation

For the initial workforce analytics prototype, CatBoost or LightGBM can be considered for structured attrition risk prediction, Random Survival Forest for time-to-attrition analysis, HDBSCAN for talent segmentation, and Isolation Forest for anomaly detection. E5/BGE or Sentence-BERT embeddings can support skill and policy retrieval, while a RAG-based conversational assistant can provide evidence-grounded workforce queries and recommendations. This combination supports predictive insights, skill-gap analysis, workforce health monitoring, learning recommendations, and natural-language interaction.

Conclusion

The proposed models broaden the analytical capabilities of the AI-Powered Workforce Analytics & Talent Intelligence Dashboard beyond traditional classification. Tree-based ensemble models strengthen attrition and retention prediction, survival models support time-to-exit analysis, clustering techniques identify meaningful talent segments, and anomaly detection methods highlight unusual workforce patterns. Transformer-based NLP models and semantic embeddings improve skill intelligence, document analysis, and workforce search, while Retrieval-Augmented Generation enables evidence-grounded conversational analytics. Together, these technologies provide a scalable and intelligent foundation for modern workforce planning and talent decision-making. Expected performance values are indicative and should be validated on the final workforce dataset.